

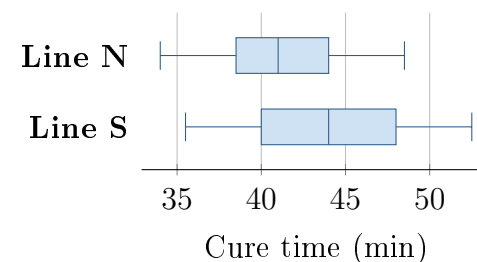
Two Test Plans for Cure Time

Unit 4 · Topic 4.9 · TODAY'S PROBLEM

Suppose line N uses a new formula for sealant, a material that closes gaps. Line S uses the standard formula. A team takes independent simple random samples of 18 of 600 recent N batches and 18 of 540 recent S batches. Mean cure times are 41.2 and 44.0 minutes; the boxplots are shown. Before sampling, the team asked, “Do the two populations differ in mean cure time?” Formula was not randomly assigned to lines.

Rhea: “Keep the original question and use population means. A difference between lines may have other causes.”

Sol: “Since $41.2 < 44.0$, use a lower tail with sample means. A significant result proves the formula caused faster curing.”



FIRST TAKE (5 min, on your sheet)

Did the team originally ask whether N was faster or whether the lines differed in either direction? Explain in one sentence.

DOK 1 (a) Define μ_N and μ_S in context.

DOK 2 (b) State H_0 and H_a in N-minus-S order to match the question asked before sampling.

DOK 3 (c) In 3–4 sentences, judge the two plans. Explain why the original question sets the tail, why hypotheses use population means, and why these two lines cannot isolate the formula as the cause.

Video + follow-along: [u7_lesson8_live.html](https://www.illustrativemathematics.org/HS/index.html) (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

